

Failure (force the parser to actually fail) — Hints

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Only 2 hints — work through them in order, and don't jump ahead until you've genuinely tried. Each hint has 2 depth levels: **Basic** (the plain idea) and **Intermediate** (the exact classes and mechanism). Read Basic first even if you're comfortable with Pydantic — it's the fastest way to spot exactly what Intermediate adds.

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Hint 1 — Forcing a real mismatch, and finding the exact error class

Basic Version

This is the previous exercise's failure case, but on purpose and closely inspected. You need a Pydantic shape with a field that's genuinely hard to fill from a certain kind of question — like a number field, asked about something that isn't a number at all.

Run the chain, catch whatever error comes back, and don't just print it — print its exact Python type. That's the part your code would need to `except` in real life.

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Intermediate Version

The goal here is to identify the *exact* exception class `.with_structured_output(...)` raises when the model's reply can't be coerced into your Pydantic shape, so your production code can catch that specific class — not a broad `Exception`.

Design a Pydantic model with a required numeric field (`int` or `float`), and ask about something purely descriptive that has no clean number in it. This forces a genuine mismatch, rather than relying on the model happening to misbehave.

Here are the exact pieces you need: - A Pydantic model with a number field: `class`

`Rating(BaseModel): score: int` - A structured chain:

`ChatOpenAI().with_structured_output(Rating)` - A question with no number in it: "Describe your favorite color." - Catch it, then print `type(e)` and `type(e).__mro__` (its full class hierarchy) — this tells you exactly which `except SomeSpecificError:` clause would catch it, and which broader classes it also happens to be an instance of.

Run this and write down the exact class name before moving to Hint 2.

A required `int` field combined with a prompt that has no number-shaped answer at all is a reliable way to force a mismatch — an *optional* field or a vaguely-numeric prompt might let the model guess something that technically parses, and you'd get a flaky test instead of a reliable one. Keep a broader `except Exception` fallback alongside the specific catch, logging `type(e).__name__` — library upgrades can change exactly which class gets raised, so a logged fallback catches that drift instead of silently missing it.

Difference between Basic and Intermediate: Basic tells you to force a mismatch and inspect the error's type. Intermediate gives you the exact mechanism (`type(e).__mro__`) and the exact pieces to force it reliably — this is the depth the exercise's Solution is written at.

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Hint 2 — The plan, and almost the whole thing

Basic Version

```
define a shape with a required number field called score

build a structured chain using that shape

ask it something with no number in the answer at all

catch whatever error happens
    print exactly what kind of error it is
```

Here is almost the whole thing:

```
# structured_output_practice.py – Failure section
from pydantic import BaseModel
from langchain_openai import ChatOpenAI

class Rating(BaseModel):
    score: int

structured_model = ChatOpenAI().with_structured_output(Rating)
```

What's missing: the `try/except` call itself, printing the exact error type, and a one-line note on which `except` clause you'd actually write in real code. Add those, then check the [Solution](#).

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Intermediate Version

```
class Rating(BaseModel):
    score: int

model = ChatOpenAI(model="...", temperature=0)
structured_model = model.with_structured_output(Rating)

try:
    prompt = "Describe your favorite color in a full sentence, no numbers."
    structured_model.invoke(prompt)
except Exception as e:
    print(type(e))          # exact class
    print(type(e).__mro__)  # its full parent chain
    print(str(e))          # the actual message
```

If this particular prompt doesn't fail (the model might still invent a number), try a more clearly non-numeric prompt, or a stricter field type (like a `Literal[...]` with a fixed set of choices that clearly excludes the answer).

What's missing to finish: the actual `except <SpecificClass>:` clause once you know what to write, replacing the broad `except Exception:`. Write it yourself, then compare against the [Solution](#).

Difference between Basic and Intermediate: Basic gets you a script that demonstrates the failure and prints its type. Intermediate turns that into the actual `except <SpecificClass>:` clause once you've identified it, with a broader fallback that logs the real class caught.

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Full solution: [Show me the solution](#)